

CSE 566 Final

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I. NEURAL NETWORKS

A. Question (12 points)

Run the linear perceptron algorithm one epoch for the following data:

Feature 1	Feature 2	Class
1	1	1
1	0	1
0	1	1
0	0	0

B. Question (14 points)

Consider the alternative error function for one hidden layer multilayer perceptron:

$$E = \frac{1}{2} \sum_{i=1}^N (r_i - o_i)^2 + \alpha \sum_{i,j} w_{ij}^2 + \beta \sum_i v_i^2 \quad (1)$$

where N represents the number of instances, r_i represents the actual output, o_i represents the calculated output, w_{ij} are the weights between the inputs and the hidden units, v_i are the weights between the hidden units and the output unit. Derive the gradient descent update rules for the weights.

II. DECISION TREES

A. Question (12 points)

Propose a tree induction method where each split in each node is in the form of $a \leq x_i \leq b$.

B. Question (12 points)

Show the steps taken by the decision tree algorithm according to the following training examples.

Class	a_1	a_2
+	T	T
+	T	T
-	T	F
+	F	F
-	F	T
-	F	T

III. KERNEL MACHINES

A. Question (14 points)

Remember the optimization problem in two class classification for the optimal separating hyperplane in Kernel machines.

$$\begin{aligned} \min \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & r^T (w^T x^t + w_0) \geq 1 \end{aligned}$$

Obtain the primal L_p and dual formulation L_d for this problem.

IV. NONPARAMETRIC METHODS

A. Question (10 points)

Given the dataset,

$$\mathbf{D} = \begin{bmatrix} 0.2 & 0.2 & 0.3 & 0.3 \\ 0.1 & 0.3 & 0.2 & 0.4 \\ 0.4 & 0.4 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.7 \end{bmatrix}$$

find the nearest example to (0.3, 0.3, 0.3, 0.3).

V. COMBINING MULTIPLE LEARNERS

A. Question (14 points)

If each base-learner is independently and identically distributed and correct with probability $p > 1/2$, what is the probability that a majority vote over L classifiers gives the correct answer?

B. Question (12 points)

Consider the three learners d_1 , d_2 , and d_3 ; and the posterior probabilities found for three classes for those learners.

	C_1	C_2	C_3
d_1	0.2	0.5	0.3
d_2	0.1	0.6	0.3
d_3	0.3	0.3	0.4

Calculate the posterior values for three classes for sum, median, minimum, maximum and product rules.